INCH-POUND

A-A-52150
September 2,1994

COMMERCIAL ITEM DESCRIPTION

REFRIGERATOR-FREEZER, EXPLOSION PROOF, NON-FOOD, LABORATORY

The General Services Administration has authorized the use of this commercial item description in preference to size 13, grade A of MIL-R-43592 for all Federal Agencies (see 5.5).

1. SCOPE

1.1 Scope. This commercial item description defines technical and quality assurance requirements for one type of reach-in, condensing unit beneath the cabinet, non-food, explosion proof, laboratory refrigerator-freezer.

2. SALIENT CHARACTERISTICS

2.1 Design and construction. The explosion-proof mechanical refrigerator-freezer shall be comprised of an insulated cabinet, evaporator and an electric driven condensing unit. The refrigerator shall be a two door model, one door for the general storage compartment, and one door for the freezer storage compartment and shall be completely explosion-proof on both the interior and exterior by complying with Underwriters Laboratories, Inc. Standard No. 471, and conforming to explosion safety requirements of the National Electric Code Standard No. 70, class I, group C & D for hazardous locations.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be used in improving this document should be addressed to: Commander, U.S. Army Natick Research, Development, and Engineering Center, ATTN: SATNC-WEE, Natick, MA 01760-5018 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4110

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2.1.1 Insulated cabinet. The total refrigerator storage area shall have a minimum of 12.5-cubic feet, including a minimum of 2.9-cubic feet of freezer storage area. Maximum overall dimensions shall be 31-inches wide, 65-inches high, and 32-1/2-inches deep. The exterior of the cabinet shall be fabricated of baked enamel on carbon steel or aluminum. The carbon steel shall have a minimum thickness of 24 gauge (0.0239-inch). The aluminum shall have aluminum thickness of 0.040-inch. The interior of the cabinet shall be fabricated of aluminum, baked enamel or porcelain enamel on carbon steel or ABS plastic. The carbon steel shall have a minimum thickness of 0.040-inch. The space between the exterior and interior of the cabinet and in the doors shall be completely filled with insulation of sufficient amount so that no condensate forms on the exterior of the cabinet during the operational test. The refrigerator cabinet shall be equipped with not less than three full-width shelves. Ice cube trays shall be furnished with the refrigerator-freezer to freeze 2.8 pounds of water. Supports or shelves shall be provided for ice cube trays in the freezer compartment. Unless otherwise specified, defrosting shall be accomplished either automatically or manually.

2.1.2 Refrigeration unit. The refrigeration unit shall be completely self-contained, pre-charged and wired with a hermetically sealed compressor unit capable of maintaining the general storage compartment temperature at 38 F $\pm$ 2 F, and the freezer storage compartment temperature at -10 F or lower in an ambient temperature of 90 F $\pm$ 2 F.

2.1.2.1 Ozone depleting substances. Class 1 ozone depleting substances, in accordance with section 602 (a) of the Clean Air Act Amendments 1990, 42 USC 7671a(a) as implemented by 40 CFR 82, Protection of Stratospheric Ozone, shall not be used for refrigerant gases, or for any other purpose, or employed in any process associated with the manufacture of any component parts for this item.

2.2 Electrical characteristics. Unless otherwise specified (see 5.2), the refrigerator-freezer shall operate on 120 volt, 60 Hz, 1-phase, electrical system, and shall be proved with a 6 foot, 3 wire cord conforming to UL 62 and provided with a 3 prong grounding type plug conforming to NEMA WD-1.

2.3 Codes and standards. The refrigerator-freezer shall comply with the applicable requirements of the following (see 3.4):

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Environmental Protection Agency (EPA) Clean Air Act Amendment
42 USC 7671a(a) and 40 CFR 82
National Electric Manufacturers Association (NEMA) Standard WD-1
National Fire Protection Association (NFPA) Standard No. 70
Underwriters Laboratories Inc. Standards No. 62, and 471

2.4 Identification marking. Each explosion proof refrigerator-freezer shall be furnished with a metal identification plate permanently attached to the exterior surface of the refrigerator in a conspicuous place. Use of a permanent "stick-on" label is acceptable. Use of removable labels is prohibited. The metal plate shall contain the following information:

Contract Number
Manufacturer's Cage Code or Name, Address, and Telephone Number
Manufacturer's Model Number Designation
Unit Capacity
Type and Amount of Refrigerant required for system
Electrical power required

3. QUALITY ASSURANCE PROVISIONS

3.1 Contractor certification. The contractor shall certify, and maintain substantiating evidence, that the product offered meets the salient characteristics and requirements of this Commercial Item Description, and that conforms to the producer's own drawings, specifications, standards, and quality assurance practices, and is the same product offered for sale in the commercial market place. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

3.2 Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within the tolerances specified using conversion tables contained in the latest revision of Federal Standard No. 376 and all other requirements of this CID are met. If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch-pound units, a request shall be made to the contracting officer to determine if the product is acceptable. The contracting officer has the option of accepting or rejecting the product.

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3.3 Regulatory requirements. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

3.4 Standards compliance. Prior to approval of the first shipment, the contractor shall submit satisfactory evidence to the contracting officer or his authorized representative that the refrigerator-freezer conforms to the requirements of those standards specified in 2.3.

Standard No. 70 - National Electrical Code

3.5 End item examination. The contractor is responsible for assuring that the product offered conforms to the stated requirements. Sampling inspection shall be in accordance with MIL-STD-105. The inspection shall be S-2 and the finding of any defect shall be cause for rejection of the lot. The sample unit shall be one complete end item. The defects are as follows:

Any missing component
Any burr, sliver, sharp edge, rough die, tool, gauge, trim or grind marks
Not constructed as specified
Any component fractured, buckled, dented, punctured, or malformed
Not fabricated of materials specified
Welding not smooth and free of cracks, burn holes, or undercuts
Fastening devices missing

3.6 End item testing. One explosion-proof refrigerator-freezer from the production lot shall be tested as specified below. Any test failure shall be cause for rejection of the lot.

3.6.1 Refrigerant leakage test. Each refrigerator-freezer shall be subjected to the refrigerant leakage test. With the refrigerant system fully charged and at rest (not in the operating cycle, refrigerant pressure equalized throughout the system), the complete refrigeration system shall be scanned with an electronic leak detector calibrated and certified to detect leakage of refrigerant-12, or other fully halogenated chlorofluorocarbon refrigerant, at a rate of one-half ounce per year or less. Indications of the presence of leaking refrigerant shall constitute failure of the leakage test. This test may be administered as a part of statistical process controls associated with the manufacturing processes for the assembly of the refrigerator-freezer, and shall be in addition to any other leak

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testing of the system completed prior to final charging with refrigerant.

3.6.2 Operational test. The explosion-proof refrigerator-freezer with the doors in an open position, shall be placed in a test room with the ambient temperature controlled at 90°F ± 20°F. After soaking for 4-hours, the doors shall be closed and the refrigerator-freezer shall be operated and the interior temperatures recorded. The refrigerator compartment shall pull down to and maintain a temperature of 38°F ± 2°F in the general storage area, and the freezer compartment shall maintain a temperature of -10°F or lower, within 8 hours. During the test, compressor unit shall not cut-out on the overload. Any failure to meet these requirements constitutes failure of the test.

3.6.3 Insulation test. During the operational test, the exterior of the refrigerator shall be visibly monitored for condensation every 30-minutes. Any presence of condensation shall constitute failure of the test.

4. PACKAGING

4.1 Packaging and packing. Packaging and packing shall be commercial or export as specified (see 5.2).

4.2 Commercial packaging and packing. Each refrigerator-freezer shall be preserved, packaged and packed in accordance with ASTM D 3951.

4.3 Export packaging. Each refrigerator-freezer shall be preserved and cushioned to protect against corrosion, deterioration and physical damage during shipment and storage.

4.3.1 Export packing. Each refrigerator-freezer shall be packed in a snug-fitting, wood-cleated plywood box conforming to PPP-B-601. The contents of each box shall be cushioned, boxed and braced to prevent movement and damage during transit. Each box shall be closed and strapped in accordance with PPP-B-601.

4.4 Marking (commercial and export). Marking of shipping containers shall be in accordance with MIL-STD-129.

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5. NOTES

5.1 Intended use. The refrigerator-freezer are intended for general laboratory use in which materials to be refrigerated may be explosive, or can be used in any room with explosive atmosphere.

5.2 Ordering data. Acquisition documents must specify the following:

a. Title, number, and date of this document.
b. When electrical requirements are other than specified (see 2.2).
c. Type of packaging and packing required (see 4.1).

5.3 Sources of documents.

5.3.1 Sources of nongovernment association documents.

ASTM D 3951 - Standard Practice for Commercial Packaging, is available from:

American Society for Testing and Materials (ASTM)
1916 Race Street
Philadelphia, PA 19103-1187

EPA Clean Air Act Amendment of 42 USC 7671a(a), is available from:

Environmental Protection Agency(EPA)
401 M Street, SW
Washington, DC 20460

NEMA Standard WD-1 - General Requirements for Wiring Devices, and NEMA 5-15P plugs, are available from:

The National Electrical Manufacturers Association (NEMA)
2101 L Street, NW, Suite 300
Washington, DC 20037

NFPA Standard No. 70 - National Electric Code, is available from:

The National Fire Protection Association (NFPA)
One Batterymarch Park
Quincy, MA 02269-9101

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UL Standard No. 62 - Standard for Safety Flexible Cord and Fixture Wire, and

UL Standard No. 471 - Standard for Safety Commercial Refrigerators and Freezers, are available from:

Underwriters Laboratories, Inc. (UL)
333 Pfingsten Road
Northbrook, IL 60062

5.3.2 Source for Government documents. Copies of military and federal documents are available from:

Standardization Documents Order Desk
Building 4D
700 Robbins Avenue
Philadelphia, PA 1911-5094

5.4 Sources of supply. Manufacturers whose products are known to meet the CID requirement are as listed below. However, competition is not limited to these companies:

Puffer Hubbard
Harris Manufacturing Company
275 Aiken Road
Asheville, NC 28804

Daigger A and Company
Daigger Scientific Div
7255 Gabe Court
Manassas, VA 22110

Precision Scientific Group
GCA Corporation
3737 West Cortland Street
Chicago, IL 60647

N J C T
775 Passaic Ave
West Caldwell, NJ 07006-6493

The Jewett Refrigerator Co., Inc.
Jewett International
2-18 Letchworth Street
Buffalo, NY 14213-1098

Jordan Scientific Products
5400 Eadom Street
Philadelphia, PA 19137-1399

5.5 Supersession data. This commercial item description is used in preference to Size 13, Grade A of MIL-R-43592. Sizes 5, 6, and 9; and Grades B, C, and D, of that document have no NSN's and are no longer procured.

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MILITARY INTERESTS:

Custodians

Army - GL
Navy - MC
Air Force - 99

Review Activities

Army - MD
Navy - MS
Air Force - 82
DLA - GS

CIVIL AGENCY COORDINATING ACTIVITY:

GSA - FSS

PREPARING ACTIVITY:

Army - GL

(Project 4110-0457)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	A-A-52191	94-08-19

3. DOCUMENT TITLE: DISPENSER, CARBONATED BEVERAGE, MECHANICALLY REFRIGERATED; COUNTERTOP DISPENSING, REMOTE SYSTEM

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
U.S. Army Natick RD&E Center	(1) Commercial 508-651-5175	(2) AUTOVON /DSN 256-5175
c. ADDRESS (Include Zip Code) Commander, U.S. Army Natick RD&E Center ATTN: SATNC-WEE Natick, MA 01760-5018	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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